American Computer Science League

2025-2026 ● Contest 4: Numbrix ● Intermediate Division

PROBLEM: *Numbrix* is an *n* x *n* grid number puzzle. The objective is to fill the grid with all of the integers from 1 to n^2 inclusive by placing all missing integers, *x*, in the grid such that both *x - 1* and *x + 1* are adjacent to it, in the same row or the same column.

In this problem, you will be given the grid size, *n*, and the integers 1 through n^2 that fill the grid in row-major order. All missing numbers are represented by 0. Your task is to determine the missing numbers and output them in row-major order.

We guarantee that:

- Two missing numbers will never be in diagonal positions. In the example below, 23 and 25 cannot both be 0.
- For every missing number, the previous number and the next number are given in the grid.
- The numbers 1 and n^2 will never be one of the missing numbers.
- There will be no more than *2n* missing numbers.

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EXAMPLE:

<table>
<tr><td>Input</td><td>5
17 0 15 12 0 18 0 14 13 10 21 20 3 4 9 22 23 2 0 0 25 0 1 6 7</td></tr>
<tr><td>Output</td><td>16 11 19 5 8 24</td></tr>
<tr><td>Explanation</td><td>The puzzle is represented in the 2-dimensional array as follows:
<table>
<tr><td>17</td><td>0</td><td>15</td><td>12</td><td>0</td></tr>
<tr><td>18</td><td>0</td><td>14</td><td>13</td><td>10</td></tr>
<tr><td>21</td><td>20</td><td>3</td><td>4</td><td>9</td></tr>
<tr><td>22</td><td>23</td><td>2</td><td>0</td><td>0</td></tr>
<tr><td>25</td><td>0</td><td>1</td><td>6</td><td>7</td></tr>
</table>
The missing integers from the grid are 5, 8, 11, 16, 19, 24. They are placed as follows:
<table>
<tr><td>17</td><td>16</td><td>15</td><td>12</td><td>11</td></tr>
<tr><td>18</td><td>19</td><td>14</td><td>13</td><td>10</td></tr>
<tr><td>21</td><td>20</td><td>3</td><td>4</td><td>9</td></tr>
<tr><td>22</td><td>23</td><td>2</td><td>5</td><td>8</td></tr>
<tr><td>25</td><td>24</td><td>1</td><td>6</td><td>7</td></tr>
</table>
Output these in row-major order: 16 11 19 5 8 24</td></tr>
</table>

INPUT: Input an integer, *n*, for the size of the puzzle and a string containing all of the integers from 1 to n^2 in an *n* x *n* two-dimensional array, using a 0 for any integer that is missing. The value of *n* will be between 4 and 9, inclusive.

OUTPUT: Output a string of integers representing the missing number(s), in row-major order, each separated by a single space.

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SAMPLE INPUT:

1. 5
   17 0 15 12 0 18 0 14 13 10 21 20 3 4 9 22 23 2 0 0 25 0 1 6 7
2. 4
   1 10 0 12 0 9 8 13 3 6 7 0 4 0 16 15
3. 6
   0 27 26 23 22 0 29 30 0 24 19 20 32 31 36 1 18 0 33 0 35 2 15 16 6 0 4 0 14 0 7 0 9 0 11 12
4. 6
   16 0 18 0 26 0 15 0 19 24 29 28 14 21 22 0 0 31 0 2 3 4 5 32 12 1 0 7 6 33 0 10 9 36 0 34
5. 7
   49 0 47 0 45 44 43 16 0 18 0 20 0 42 15 14 13 24 23 22 41 0 1 0 0 0 33 40 3 10 11 26 31 34 0 0 9 8 27 0 35 38 5 6 0 28 29 36 0
6. 8
   60 61 62 0 50 43 0 41 0 64 63 52 49 44 45 40 58 55 0 53 0 47 46 0 0 56 13 12 11 10 9 38 18 17 14 1 0 7 0 0 0 16 15 2 5 30 31 36 20 23 24 3 0 0 32 35 0 22 0 26 27 28 33 0

SAMPLE OUTPUT:

1. 16 11 19 5 8 24
2. 11 2 14 5
3. 28 21 25 17 34 5 3 13 8 10
4. 17 25 27 20 23 30 13 8 11 35
5. 48 46 17 19 21 2 12 25 32 39 4 30 7 37
6. 51 42 59 54 48 39 57 6 8 37 19 4 29 21 25 34

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TEST DATA

TEST INPUT:

1. 4
   10 0 8 0 11 12 5 6 14 13 0 3 0 16 1 2
2. 5
   0 2 1 10 11 4 7 0 9 0 0 6 15 14 13 18 17 0 23 0 19 20 0 22 25
3. 6
   17 0 23 0 27 0 16 19 22 25 26 29 0 20 21 34 0 30 14 1 36 35 32 31 13 2 0 4 5 6 0 11 10 9 0 7
4. 7
   47 0 49 0 41 36 35 46 45 44 43 40 37 0 11 12 15 16 39 38 33 10 0 14 17 0 19 32 9 8 23 22 21 20 31 6 7 24 0 26 0 30 5 0 3 2 1 28 29
5. 8
   23 0 19 0 3 0 5 6 24 21 20 17 2 9 8 7 0 26 15 0 1 10 45 0 28 27 14 13 12 0 44 47 29 0 31 0 35 42 43 0 62 0 32 33 36 41 40 49 61 64 57 56 0 38 0 50 0 59 58 55 0 53 0 51
6. 9
   65 64 0 62 0 60 0 58 57 0 67 36 37 38 39 40 55 0 69 68 0 34 0 24 0 54 53 0 31 32 33 26 23 42 43 52 71 30 29 0 27 22 0 44 0 72 79 80 81 20 21 46 47 50 73 78 0 18 0 4 5 48 49 0 77 16 1 2 3 6 0 8 75 76 15 0 13 12 11 10 9

TEST OUTPUT:

1. 9 7 4 15
2. 3 8 12 5 16 24 21
3. 18 24 28 15 33 3 12 8
4. 48 42 34 13 18 25 27 4
5. 22 18 4 25 16 46 11 30 34 48 63 37 39 60 54 52
6. 63 61 59 66 56 35 25 41 70 28 45 51 17 19 74 7 14

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问题: *Numbrix* 是一个 n x n 的网格数字谜题。目标是将从 1 到 n^2 的所有整数填入网格，即把所有缺失的整数 x 放入网格中，使得 $x - 1$ 和 $x + 1$ 分别与 x 在同一行或同一列相邻。

在本题中，已知网格的大小 n ，以及按行优先顺序填充网格的整数 1 到 n^2。所有缺失的数字用 0 表示。你的任务是确定缺失的数字，并按行优先顺序输出它们。

我们需确保：

- **两个缺失的数字绝不会出现在对角位置。在下面的例子中，**23 **和** 25 **不能同时为** 0**。**
- **对于每个缺失数字，其前一个数和后一个数均已在网格中给出。**
- **数字** 1 **和** n^2 **绝不会是缺失数字。**
- **缺失数字的数量不超过** $2n$**。**

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示例：

<table>
<tr><td>输入</td><td>5
17 0 15 12 0 18 0 14 13 10 21 20 3 4 9 22 23 2 0 0 25 0 1 6 7</td></tr>
<tr><td>输出</td><td>16 11 19 5 8 24</td></tr>
<tr><td>解析</td><td>该谜题在二维数组中的表示方式如下：
<table>
<tr><td>17</td><td>0</td><td>15</td><td>12</td><td>0</td></tr>
<tr><td>18</td><td>0</td><td>14</td><td>13</td><td>10</td></tr>
<tr><td>21</td><td>20</td><td>3</td><td>4</td><td>9</td></tr>
<tr><td>22</td><td>23</td><td>2</td><td>0</td><td>0</td></tr>
<tr><td>25</td><td>0</td><td>1</td><td>6</td><td>7</td></tr>
</table>
网格中缺失的整数是 5, 8, 11, 16, 19, 24。它们的位置如下所示：
<table>
<tr><td>17</td><td>16</td><td>15</td><td>12</td><td>11</td></tr>
<tr><td>18</td><td>19</td><td>14</td><td>13</td><td>10</td></tr>
<tr><td>21</td><td>20</td><td>3</td><td>4</td><td>9</td></tr>
<tr><td>22</td><td>23</td><td>2</td><td>5</td><td>8</td></tr>
<tr><td>25</td><td>24</td><td>1</td><td>6</td><td>7</td></tr>
</table>
按行优先顺序输出这些整数: 16 11 19 5 8 24</td></tr>
</table>

输入：输入一个整数 n 表示网格的尺寸，以及一个字符串，该字符串中包含 n x n 二维数组中的 1 到 n^2 所有整数，缺失的数字用 0 表示。n 的取值范围是 4 到 9（包含两端）。

输出：输出一个由整数组成的字符串，表示按行优先顺序排列的缺失数字，每个数字之间用单个空格分隔。

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样例输入：

1. 5
   17 0 15 12 0 18 0 14 13 10 21 20 3 4 9 22 23 2 0 0 25 0 1 6 7
2. 4
   1 10 0 12 0 9 8 13 3 6 7 0 4 0 16 15
3. 6
   0 27 26 23 22 0 29 30 0 24 19 20 32 31 36 1 18 0 33 0 35 2 15 16 6 0 4 0 14 0 7 0 9 0 11 12
4. 6
   16 0 18 0 26 0 15 0 19 24 29 28 14 21 22 0 0 31 0 2 3 4 5 32 12 1 0 7 6 33 0 10 9 36 0 34
5. 7
   49 0 47 0 45 44 43 16 0 18 0 20 0 42 15 14 13 24 23 22 41 0 1 0 0 0 33 40 3 10 11 26 31 34 0 0 9 8 27 0 35 38 5 6 0 28 29 36 0
6. 8
   60 61 62 0 50 43 0 41 0 64 63 52 49 44 45 40 58 55 0 53 0 47 46 0 0 56 13 12 11 10 9 38 18 17 14 1 0 7 0 0 0 16 15 2 5 30 31 36 20 23 24 3 0 0 32 35 0 22 0 26 27 28 33 0

样例输出：

1. 16 11 19 5 8 24
2. 11 2 14 5
3. 28 21 25 17 34 5 3 13 8 10
4. 17 25 27 20 23 30 13 8 11 35
5. 48 46 17 19 21 2 12 25 32 39 4 30 7 37
6. 51 42 59 54 48 39 57 6 8 37 19 4 29 21 25 34

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测试数据

测试输入:

1. 4
   10 0 8 0 11 12 5 6 14 13 0 3 0 16 1 2
2. 5
   0 2 1 10 11 4 7 0 9 0 0 6 15 14 13 18 17 0 23 0 19 20 0 22 25
3. 6
   17 0 23 0 27 0 16 19 22 25 26 29 0 20 21 34 0 30 14 1 36 35 32 31 13 2 0 4 5 6 0 11 10 9 0 7
4. 7
   47 0 49 0 41 36 35 46 45 44 43 40 37 0 11 12 15 16 39 38 33 10 0 14 17 0 19 32 9 8 23 22 21 20 31 6 7 24 0 26 0 30 5 0 3 2 1 28 29
5. 8
   23 0 19 0 3 0 5 6 24 21 20 17 2 9 8 7 0 26 15 0 1 10 45 0 28 27 14 13 12 0 44 47 29 0 31 0 35 42 43 0 62 0 32 33 36 41 40 49 61 64 57 56 0 38 0 50 0 59 58 55 0 53 0 51
6. 9
   65 64 0 62 0 60 0 58 57 0 67 36 37 38 39 40 55 0 69 68 0 34 0 24 0 54 53 0 31 32 33 26 23 42 43 52 71 30 29 0 27 22 0 44 0 72 79 80 81 20 21 46 47 50 73 78 0 18 0 4 5 48 49 0 77 16 1 2 3 6 0 8 75 76 15 0 13 12 11 10 9

测试输出：

1. 9 7 4 15
2. 3 8 12 5 16 24 21
3. 18 24 28 15 33 3 12 8
4. 48 42 34 13 18 25 27 4
5. 22 18 4 25 16 46 11 30 34 48 63 37 39 60 54 52
6. 63 61 59 66 56 35 25 41 70 28 45 51 17 19 74 7 14